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ARTICLE

Genetic Assignment of Brook Trout Reveals Rapid Success of Culvert Restoration in Headwater Streams

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Abstract

Fragmentation of aquatic habitats is a global conservation concern. Losses of both biodiversity and genetic diversity have been attributed to this phenomenon as dispersal movements between habitat patches are restricted. Brook Trout *Salvelinus fontinalis* in the upper Shavers Fork watershed of West Virginia exhibit high dispersal rates and lengthy dispersal distances; however, road culverts have been hypothesized to influence the movement of Brook Trout between tributaries. Removing impassable culverts could serve as a tool to facilitate movements into previously isolated streams. We classified the passability of road culverts by their physical attributes and collected samples above each structure as well as in two streams without culverts to test for genetic population differentiation attributed to culverts. After we determined that the majority of streams consisted of a single genetic population, with the exception of two streams containing culverts designated as impassable (Beaver Creek and Lamothe Hollow), the culvert structures were restored to eliminate outlets perched above the stream bottom and were subsequently resampled 1 year after project completion. Although no individuals were assigned as first-generation migrants into either Beaver Creek or Lamothe Hollow before culvert restoration, five individuals were detected as first-generation migrants into Beaver Creek (13%), and one individual was identified as a first-generation migrant into Lamothe Hollow (4.5%) after culvert restoration. Three different streams were found to be sources of the migrants into Beaver Creek, and the single individual migrating into Lamothe Hollow was from a fourth source stream. The identification of first-generation migrants through genetic assignment not only provides a tool with which to test the effectiveness of barrier removal and subsequent restoration success but also to support the importance of culvert replacement in headwater streams.

Fragmentation of natural habitats occurs through the transformation of a single continuous habitat into an increased number of smaller, distinct, isolated patches. Although fragmentation occurs naturally across diverse landscapes (e.g., fire: Wright 1974; waterfalls: Dias et al. 2013), human land use practices have contributed to the global subdivision of large-scale, continuous systems (Haddad et al. 2015), resulting in losses of both biodiversity (e.g., Gibson et al. 2013) and genetic diversity (e.g., Dixon et al. 2007) as well as an increased risk for subpopulation extinction (Letcher et al. 2007). Habitat fragmentation is especially a concern within linear aquatic systems (Fagan 2002), as individuals in these systems often disperse outside their natal range in response to spatiotemporal variability in chemical, physical, and biological states (Hildrew and Giller 1994; Petty et al. 2012).

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Road culverts have been identified as sources of habitat fragmentation and artificial movement barriers in headwater streams (Warren and Pardew 1998; Torterotot et al. 2014) due to steep interior slopes (Poplar-Jeffers et al. 2009), insufficient water depth (Gibson et al. 2005), high outlet drop (Riley 2004; Burford et al. 2009; Poplar-Jeffers et al. 2009; MacPherson et al. 2012), increased water velocities (Belford and Gould 1989; Gibson et al. 2005), and culvert length (Bouska and Paukert 2009; Briggs and Galarowicz 2013). Although prior studies of culverts have noted varying levels of passability among culverts (Poplar-Jeffers et al. 2009), consequences of impassable culverts in other linear systems include reduced species composition (Warren and Pardew 1998; Vander Pluym et al. 2008; Nislow et al. 2011), genetic differentiation (Torterotot et al. 2014), and loss of genetic diversity (Wofford et al. 2005; Neville et al. 2009).

Brook Trout *Salvelinus fontinalis* in the central Appalachian Mountains are well-studied headwater stream specialists and serve as sentinels for conditions in coldwater systems. Brook Trout populations have experienced a significant and continuing reduction in their historical native range (eastern United States) over the past century as a result of competition with stocked salmonids (i.e., Brown Trout *Salmo trutta* and Rainbow Trout *Oncorhynchus mykiss*), acid precipitation, climate change, and habitat degradation (Petty and Thorne 2005; Flebbe et al. 2006; McClurg et al. 2007; Hudy et al. 2008; Richards et al. 2008). As a consequence, most current populations within the Appalachian region exist as spatially isolated and genetically differentiated subpopulations (Richards et al. 2008; Hudy et al. 2010; Kanno et al. 2011; Whiteley et al. 2013; Aunins et al. 2014). The resulting isolation can increase the likelihood of genetic drift and susceptibility to stochastic extinction through the loss of genetic variability (Fagan 2002; Allendorf and Luikart 2007; Whiteley et al. 2013).

In contrast to Brook Trout in other locations within the species' range, Brook Trout in the Shavers Fork watershed appear to have high levels of dispersal among streams, a life history strategy to maximize both growth and reproductive success (Rogers and Curry 2004; Stolarski and Hartman 2010; Petty et al. 2005, 2012, 2014; Utz and Hartman 2006). Movement studies within this watershed have shown that Brook Trout disperse seasonally between tributaries by way of a coldwater main stem, resulting in a dendritic network of highly mobile Brook Trout populations (Petty et al. 2005, 2012, 2014). Additionally, while most current populations within the Appalachian region exist as spatially isolated and genetically differentiated subpopulations (Richards et al. 2008; Hudy et al. 2010; Kanno et al. 2011; Whiteley et al. 2013; Aunins et al. 2014), a genetics study by Aunins et al. (2014) suggested that in the absence of barriers, Brook Trout within the Shavers Fork watershed demonstrated a high degree of gene flow among the subpopulations; those authors concluded that the relatively coldwater main stem provided a high-quality corridor for dispersal among small streams. This connectivity among unrestricted streams makes Brook Trout in the Shavers Fork the ideal candidate species for studying the effects of barriers on movement.

Although Brook Trout in Shavers Fork appear to naturally disperse among streams through the coldwater main stem, culverts with varying levels of passability occur in a number of headwater streams as a result of a historic railroad (installed before 1910) paralleling the Shaver's Fork as well as dirt and gravel roads in the watershed. While losses in both species (e.g., Warren and Pardew 1998) and genetic diversity (Wofford et al. 2005) of in-stream fishes have been widely attributed to road culverts, passability of individual culverts may vary across the landscape (Poplar-Jeffers et al. 2009). Assessments of culvert passability have utilized both biological indices (including mark and recapture; e.g., Vander Pluym et al. 2008) and population characteristics (e.g., species richness; Nislow et al. 2011), but the results of these types of studies can be inconclusive due to low recapture numbers (Bouska and Paukert 2009) and temporal effects (e.g., migration; Blank et al. 2005). Software, including FishXing version 2.2, has been used to evaluate the passability of culverts based on their physical features. However, several studies have noted differences between field assessments and estimated passability by FishXing 2.2 (Blank et al. 2005; Bourne et al. 2011). Additionally, Poplar-Jeffers et al. (2009) found that FishXing 2.2 was too exclusive by deeming all 120 culverts surveyed impassable. Of the same streams, the Love and Taylor (2003) phase I protocol identified less than 70% of these structures as impassable, allowing for prioritization of culvert improvements (e.g., baffles) or restoration for culverts classified as impassable (Poplar-Jeffers et al. 2009).

In addition to assessing culverts' physical features to determine passability, genetic analyses have also been used to assess passability through indices of genetic diversity and differentiation (e.g., Torterotot et al. 2014). Unlike mark–recapture techniques that assess an individual's ability to traverse a potential barrier, measures of genetic diversity, including observed heterozygosity (H_o) and genetic differentiation (e.g., F_{ST}; Wright 1951), account for a population's movement ability to evaluate barrier passability through genetic comparisons with other populations (Goerig et al. 2015). Additionally, genetic analyses also provide an opportunity to evaluate individual-based movements through genetic assignment. Evaluations of genetic assignment found a high level of accuracy of assignment to source populations (Berry et al. 2004), therefore providing an opportunity to assess dispersal of first-generation migrants into restored headwater streams after culvert restoration. Although restoring culverts may increase species richness (Franklin and Bartels 2012;

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Evans et al. 2015), connecting potentially isolated populations into a continuous riverscape is viewed as a strategy to remedy possible losses in genetic diversity within disconnected subpopulations (Fausch et al. 2002; Letcher et al. 2007) and to maintain the evolutionary potential of populations (Lowe and Allendorf 2010). The objectives of this study were to (1) determine whether purportedly impassable culverts (sensu Poplar-Jeffers et al. 2009 and based on the Love and Taylor 2003 phase I protocol) resulted in genetic isolation and loss of genetic diversity in the upstream population and (2) assess whether replacement of impassable culverts resulted in migration from other populations 1 year after culvert restoration.

METHODS

Study area.— The upper Shavers Fork is a large (>150-km^2), high-elevation (originating at 1,500 m) watershed located in the Monongahela National Forest of eastern West Virginia (Figure 1). Land cover of the area is dominated by a mixed deciduous–coniferous forest, resulting in a dense tree canopy within tributaries. Unique to Shavers Fork is a coldwater main stem (temperatures rarely exceed 25°C; Petty et al. 2012), allowing for spatial linkage between populations inhabiting the smaller tributaries. Brook Trout residing in the Shavers Fork main stem exhibit high daily rates of movement (up to 50 m/d; Petty et al. 2012); however, abiotic factors, including a lack of

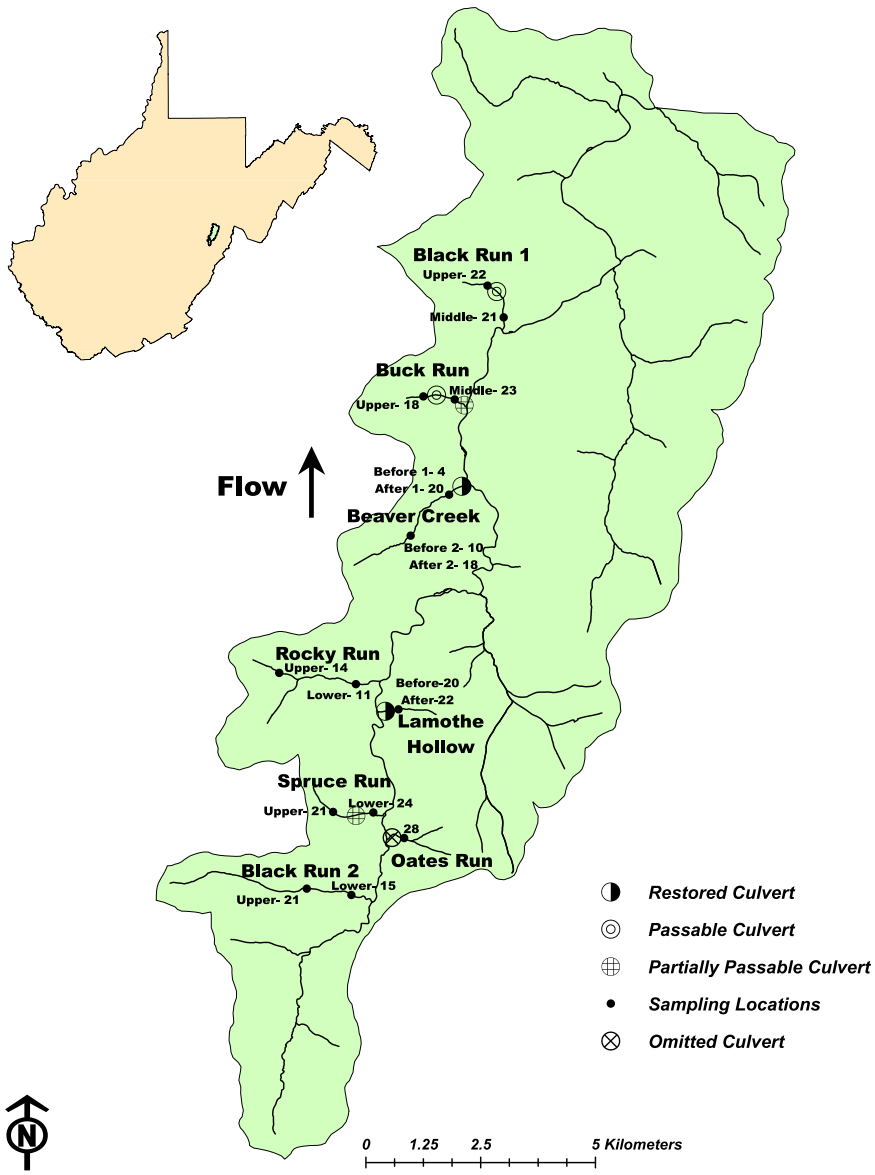

FIGURE 1. Sampling locations in the upper Shavers Fork watershed, eastern West Virginia. Classification of culverts followed the Love and Taylor (2003) phase I protocol. Numbers next to the sampling site indicate the total number of Brook Trout sampled (not including age-0 individuals). [Color figure can be viewed at afsjournals.org.]

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suitable substrate, prohibit Brook Trout from spawning in the main stem and require them to return to smaller (<3-km^2), alkaline headwater streams (Petty et al. 2005).

Early 20th-century commercialization of abundant red spruce *Picea rubens* brought about dramatic changes to the area, including the construction of a railroad paralleling the edge of the Shavers Fork main stem. Due to the large number of small headwater streams, multiple culverts were installed under the railroad. However, as a result of the steep gradient, many of the culverts were installed with the structure hanging above the bottom of the streambed (high outlet hang). In an effort to restore connectivity, a shotgun culvert (outlet hang > 1 m) was removed at Beaver Creek during the summer of 2011 (Supplemental Figure S1 available in the online version of this article) and at Lamothe Hollow during the summer of 2013 (Figure S2). To avoid outlet hang at both sites, a series of angled log weirs created a steplike series of pools, providing gradual access to the culverted stream sections. Restoration of Beaver Creek included the placement of continuous substrate within the culvert to provide instream habitat and variable water velocities (Figure S3), whereas baffles were used inside the culvert at Lamothe Hollow (Figure S4).

Culvert surveying and classification.— For all culverted stream sites, including Beaver Creek and Lamothe Hollow before restoration, a Leica Geosystems laser and rod were used to complete a longitudinal profile of the stream beginning from the tailwater control of the first resting habitat upstream of the culvert inlet to the tailwater control of the plunge pool below the culvert outlet. In addition, a cross-sectional survey was conducted at the maximum depth of the plunge pool below the culvert outlet. After the survey, classification of each culvert followed the Love and Taylor (2003) phase I protocol, which filters culverts into restoration prioritization groups through passability parameters, including channel width versus inlet width, plunge pool depth, outlet drop, interior water depth, culvert slope, and improvements to current culverts, including baffles and weirs. According to the Love and Taylor (2003) phase I protocol, culverts classified as impassable have outlet drops greater than 0.6 m or have slopes greater than 3% without baffles or weirs. Culverts classified as partially passable have an outlet drop less than 0.6 m; have slopes less than 3% without baffles or weirs; or have a slope greater than 3% but have baffles or weirs. Culverts with a continuous substrate throughout the structure and that have no outlet drop are deemed passable; those that do not have substrate but have a water depth greater than 0.15 m at both the culvert inlet and outlet are also deemed passable. Brook Trout in laboratory conditions have been observed jumping 0.735 m (Kondratieff and Myrick 2006), but a small percentage increase in culvert slope, especially exceeding 3%, greatly decreases the probability of Brook Trout passability (Goerig et al. 2015).

Fish sampling and tissue collection.— Fourteen locations with Brook Trout populations were sampled, consisting of second- and third-order tributaries of the Shavers Fork main stem, during June and July of 2012 and 2013 (Figure 1). Two sites in Beaver Creek were sampled above the purportedly impassable culvert before restoration (June 2011) and after restoration (June 2012). Additionally, Lamothe Hollow was sampled before restoration (June 2012) and after restoration (June 2014; Table 1). Culvert restoration in Lamothe Hollow occurred in June 2013. For all sampled sites, Brook Trout consisting of multiple cohorts (classified by TL; Petty et al. 2005) were collected via backpack electrofishing (Model LR-24; Smith-Root, Vancouver, Washington) in a 150-m single upstream pass and were anesthetized using clove oil. Tissue was taken from the adipose and caudal fins and was preserved in 95% ethanol until extraction.

Genetic protocols.— For all tissue sample collections, genomic DNA was extracted using the Wizard SV-96 DNA purification system (Promega, Madison, Wisconsin) according to manufacturer's protocol. Extracted DNA was quantified using a spectrophotometer (NanoDrop, Wilmington, Delaware) and was standardized to a concentration of 10 ng/μL. A suite of 13 microsatellite loci (described by King et al. 2012) was amplified using either an MJ Research PTC-200 thermocycler or a BioRad C1000 thermocycler using 10-μL reactions consisting of 1 × Promega FlexiBuffer, 1.5-mM $MgCl_2$, 0.2-mM deoxynucleotide triphosphates, 0.2-μM forward primer, 0.2-μM reverse primer, 2 units of GoTaq DNA polymerase, and 20 ng of DNA.

The amplification protocol for loci *SfoB52*, *SfoC79*, *SfoD100*, *SfoC24*, *SfoC28*, *SfoC115*, and *SfoC113* started with an initial heating of 94°C and then continued with 35 cycles of 94°C (30 s), 56°C (30 s), and 72°C (45 s), with a final hold of 72°C for 10 min (King et al. 2012). Loci *SfoC86*, *SfoD91*, *SfoC38*, *SfoD100*, *SfoC88*, and *SfoC129* were initially heated at 94°C and continued with 15 cycles of 94°C (45 s), 60°C (45 s) with a decrease of 0.5°C per cycle, and 72°C (30 s). The protocol continued for an additional 15 cycles of 94°C (45 s), 52°C (45 s), and 72°C (30 s). Resultant fragments were genotyped using the GenomeLab GeXP genetic analysis system (Beckman Coulter, Brea, California) and were scored by a single individual.

Genetic analysis.— Each of the 14 different site locations, as well as the three postrestoration collections, was treated as an individual sampling (N = 17). Possible genotyping errors or null alleles were confirmed through MICRO-CHECKER version 2.2.3 (Van Oosterhout et al. 2004). Deviations from Hardy–Weinberg equilibrium (HWE) were tested in GENEPOP (Raymond and Rousset

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TABLE 1. Comparisons of Brook Trout genetic diversity following grouping of impassable culverts and all other sites (excluding postrestoration sites). Indices of genetic diversity (H_e = expected heterozygosity; H_o = observed heterozygosity; A_R = allelic richness) were calculated in FSTAT version 2.9.3 (Goudet 2001) and Arlequin version 3.5 (Excoffier and Lischer 2010). Allelic richness was corrected for the smallest population size. Culverts were classified through the Love and Taylor (2003) phase I protocol.

Sampling site	N	H_e	H_o	A_R	Culvert below sample site	Culvert classification	Basis of classification
Buck Run Upper	18	0.52	0.53	2.86	Yes	Passable	Substrate throughout culvert; no outlet drop
Buck Run Middle	23	0.57	0.58	2.87	Yes	Partially passable	Outlet drop ≤ 0.6 m; no slope
Black Run 1 Upper	22	0.59	0.58	3.01	Yes	Passable	Substrate throughout culvert; no outlet drop
Black Run 1 Middle	21	0.50	0.51	2.86	No		
Upper Rocky	14	0.56	0.60	2.97	No		
Lower Rocky	11	0.45	0.51	2.83	No		
Oates Run	28	0.43	0.51	2.81	Yes	Impassable[a]	Slope > 3%; no baffles or weirs present
Spruce Lower	24	0.52	0.52	2.88	No		
Spruce Upper	21	0.50	0.53	2.66	Yes	Partially passable	Outlet drop ≤ 0.6 m; no slope
Black Run 2 Lower	15	0.54	0.54	3.04	No		
Black Run 2 Upper	21	0.54	0.57	2.87	No		
Lamothe Hollow	20	0.59	0.57	2.72	Yes	Impassable	Outlet drop > 0.6 m
Lamothe Hollow after restoration	22	0.41	0.48	2.40	Yes	Passable	Substrate throughout culvert; inlet width ≥ active channel width
Beaver Creek 1	4	0.44	0.59	2.80	Yes	Impassable	Outlet drop > 0.6 m
Beaver Creek 1 after restoration	20	0.56	0.57	2.72	Yes	Partially passable	Substrate throughout culvert; inlet width ≥ active channel width
Beaver Creek 2	10	0.53	0.54	2.77	Yes	Impassable	Outlet drop > 0.6 m
Beaver Creek 2 after restoration	18	0.58	0.54	2.60	Yes	Partially passable	Substrate throughout culvert; inlet width ≥ active channel width

[a]Culvert was removed from genetic diversity comparisons due to recent installation.

1995) using 10,000 iterations, and significance was determined after a sequential Bonferroni correction (Rice 1989).

Genetic analysis: assessment of genetic diversity and differentiation due to culverts.— Genetic diversity at impassable culvert sites prior to restoration (Beaver Creek 1 and 2 before restoration [combined]; and Lamothe Hollow before restoration) was compared to genetic diversity at all other sites, including sites above passable culverts, by using the program FSTAT version 2.9.3 (Goudet 2001) with significance based on 1,000 permutations. Observed heterozygosity was calculated in Arlequin version 3.5 (Excoffier and Lischer 2010), and allelic richness (A_R) and average expected heterozygosity (H_e) within groups were calculated in FSTAT version 2.9.3 (Goudet 2001). Significance of genetic differences was based on 1,000 permutations. Estimates of the effective population size (N_e) were calculated in NeEstimator version 2 (Do et al. 2014).

Genetic differentiation between all sampling locations, measured as F_{ST} (Wright 1951), was calculated using FSTAT version 2.9.3 (Goudet 2001), and the significance of each F_{ST} value was corrected for multiple comparisons using a sequential Bonferroni correction (Rice 1989). Due to the small number of individuals in the Beaver Creek 2 site before restoration ($n = 4$), the sample site was removed from F_{ST} analysis. To determine whether the suite of microsatellites could detect varying population differentiation ($F_{ST} = 0.005$ and 0.01) at different N_e values ($N_e = 100$ and 500), 1,000 independent simulations were run in the program POWSIM (Ryman and Palm 2006) using the number of alleles, allelic frequency, and sample sizes from our data set.

To determine the most likely number of populations (K), including the possibility of more genetic populations present than samplings ($N = 17$), the Bayesian software STRUCTURE (Pritchard et al. 2000) was used to determine likelihoods of possible K-values from 1 to 19 (i.e., sampling locations + 2) using both the admixture and nonadmixture models and the sampling locations as priors. Ten iterations of possible values (100,000 burn-ins and 100,000 Markov chain–Monte Carlo repetitions) were used, and the results were imported into STRUCTURE HARVESTER (Earl and vonHoldt 2012) to visually determine the most likely value of K by examining the $\log_e$ likelihood of the

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number of populations and ΔK (Evanno et al. 2005). Since STRUCTURE determines the deepest level of genetic differentiation, an iterative approach was taken to further define the population structure of each cluster at a finer scale (Vähä et al. 2007). Each grouping of K from the original determination was re-run separately through STRUCTURE using the same parameters until K was equal to 1.

Genetic analysis: detection of first-generation migrants after culvert replacement.— To identify potential first-generation migrants into Beaver Creek 1, Beaver Creek 2, and Lamothe Hollow after culvert restoration, two approaches in GENECLASS version 2.0 (Piry et al. 2004) were used, including the Bayesian-based method described by Rannala and Mountain (1997) and a frequency-based method described by Paetkau et al. (1995). The Rannala and Mountain (1997) method identifies first-generation migrants by calculating the probability that an individual's genotype is observed given the density of allele frequencies in each population. The Paetkau et al. (1995) method calculates the expected frequency of each individual given the allele density in each population and assigns based upon the highest frequency. The Bayesian likelihood algorithm from Paetkau et al. (2004) was used to generate critical values for detecting first-generation migrants by using 1,000 simulated individuals. A type I error rate of 0.05 was used to reject the null hypothesis that an individual was born to the population from which it was sampled. All tributary populations, including Beaver Creek 1 and 2 and Lamothe Hollow before culvert replacement, were used as reference populations. Individuals detected as first-generation migrants were assigned to source streams by using the highest likelihood (−log; Paetkau et al. 2004).

Brook Trout migrants into Beaver Creek 1, Beaver Creek 2, and Lamothe Hollow after culvert restoration were also identified based on cluster membership coefficients (Q) calculated in STRUCTURE (Pritchard et al. 2000). Individuals were classified as potential migrants if they had a Q-value of at least 0.50 for a cluster from which they were not sampled.

RESULTS

Three culverts (Beaver Creek, Lamothe Hollow, and Oates Run) were deemed impassable. The culverts at Beaver Creek (before replacement) and Lamothe Hollow were classified as impassable due to high outlet drops (>1 m), whereas the culvert at Oates Run was deemed impassable because of steep interior slopes (>3°) and a long culvert length (40 m) without the presence of baffles or weirs (Table 1). The culverts at Beaver Creek and Lamothe Hollow have been in place since the early 1900s, whereas the culvert at Oates Run was constructed in the early 1990s (P. Kinder, West Virginia University, personal communication). The number of elapsed Brook Trout generations (generation time = 2 years) between culvert installation and sampling may have been insufficient to permit detection of genetic signals related to movement barriers. Therefore, Oates Run was removed from further classification comparisons.

Two culverts, located at Buck Run (near the confluence of the main stem) and Spruce Run, were partially impassable due to small outlet drops. Although these outlet drops were significantly shorter than those at Beaver Creek and Lamothe Hollow, they still could preclude the movement of juvenile salmonids. In addition, two culverts located at Black Run Upper and Buck Run Upper were deemed completely passable to all Brook Trout life stages (Table 1). Two additional streams that did not have any culverts (Rocky Run and Black Run 2) were sampled at two different locations to control for the effects of geographic distance. Although these two control streams were not surveyed (i.e., due to the absence of a culvert), no apparent natural barriers (e.g., waterfalls or cascades) were observed during sampling.

Overall, 389 Brook Trout tissue samples were successfully genotyped from 17 different samplings (14 sampling locations; Figure 1). There were no indications of allelic dropout, genotyping errors, or null alleles. After correction for multiple comparisons, six loci had significant departures from HWE expectations. Because individuals from a single cohort can create a heterozygote deficiency due to potential relatedness, young-of-the-year (age-0) Brook Trout (n = 77) were removed from the analysis. After age-0 removal, three loci (*SfoB52*: $P = 0.0044$; *SfoD75*: $P = 0.0022$; *SfoC129*: $P = 0.0000$) did not conform to HWE expectations after sequential Bonferroni correction, and these loci were removed from further analysis. Sampling sites affecting these loci included Buck Run Middle and Black Run 1 Middle (*SfoB52*); Black Run 1 Middle and Beaver Creek 1 after restoration (*SfoD75*); and Black Run 1 Upper and Beaver Creek 1 before restoration (*SfoC129*).

Assessment of Genetic Diversity and Differentiation due to Culverts

No significant differences in A_R, H_o, or average H_e were observed between impassable culvert sites (Beaver Creek 1 and 2 before restoration [combined]; Lamothe Hollow before restoration) and all other sites excluding postrestoration samplings (Table 2), indicating no statistically significant loss of genetic diversity due to the presence of impassable culverts. Average estimates of N_e for Beaver Creek (221.8) and all other sites (211.8) with the exception of Lamothe Hollow were comparable. However, both Beaver Creek and Lamothe Hollow had estimates with wide confidence intervals (31.4 to infinity and 6.5 to infinity, respectively), whereas the confidence interval for the average (211.8) at all other sites was 119.2–529.4 (Table 2).

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TABLE 2. Comparisons of Brook Trout genetic diversity following grouping of impassable culverts and all other sites (excluding postrestoration sites). Indices of genetic diversity (A_R = allelic richness; H_o = observed heterozygosity; H_e = expected heterozygosity) were calculated in FSTAT version 2.9.3 (Goudet 2001) and Arlequin version 3.5 (Excoffier and Lischer 2010). Estimates of effective population size (N_e) were calculated in NeEstimator version 2 (Do et al. 2014).

Group	A_R	H_o	H_e	Estimate of N_e
Impassable culverts				
Beaver Creek (N = 14)	4.00	0.47	0.52	221.8 (31.4 to infinity)
Lamothe Hollow (N = 20)	3.71	0.56	0.58	23.5 (6.5 to infinity)
All other sites (N = 218; excluding postrestoration sites)	4.36	0.51	0.53	211.8 (119.2–529.4)
P-value	0.36	0.92	0.61	

Population Differentiation due to Culverts

Genetic differentiation was observed among most of the sampling sites. After performing a sequential Bonferroni correction, significant pairwise F_{ST} values were found between the majority (63%) of sampling sites, with the exception of control locations without a culvert (Lower Rocky and Black Run 2 Lower; Table 3). Excluding samplings after culvert restoration, all sampled populations were significantly differentiated from Lamothe Hollow; however, three sampled sites (Black Run Upper 1, Black Run Middle 1, and Lower Rocky) had nonsignificant pairwise F_{ST} comparisons to Beaver Creek 2 before restoration. Although we detected low levels of genetic differentiation except in populations above impassable culverts, independent simulations in POWSIM indicated that our panel of loci had significant power to detect population differentiation with our existing sample sizes (i.e., >99% probability of detecting an F_{ST} of 0.01; >89% probability of detecting an F_{ST} of 0.005).

The most likely number of populations (K) for both the admixture and non-admixture models was found initially to be 3 using the log likelihood of the number of populations ($\log_e P[K]$) and ΔK (Figure 2; Evanno et al. 2005). Individuals at Lamothe Hollow and Beaver Creek (impassable culverts) before and after culvert replacement were strongly differentiated from those at all other sites (Figure 3). Although both populations at Black Run were composed of an admixture belonging to Beaver Creek, the majority of individuals (65%) had Q-values belonging to the cluster containing populations not found above an impassable culvert and therefore were grouped with those populations. An iterative approach was taken to test for further delineation of fine-scale population structure of the upper Shavers Fork from the three groups identified from the initial STRUCTURE result using the previous methodology. When run separately, the most likely number of populations for all three groupings was $K = 1$, supporting the initial identification of $K = 3$ across all populations.

Detection of Migrants after Culvert Replacement

Using both the Bayesian-based simulation algorithm described by Rannala and Mountain (1997) and the frequency-based method described by Paetkau et al. (1995) within GENECLASS version 2.0 (Piry et al. 2004), five adult individuals (13.2%) were found to be likely first-generation migrants ($P < 0.05$) into Beaver Creek after culvert replacement and most likely originated from Buck Run Upper, Black Run 1 Upper, and Black Run 2 Upper (Table 4). Additionally, a single individual (5.0%) was detected as a first-generation migrant into Lamothe Hollow after culvert restoration, most likely dispersing from Oates Run. Although STRUCTURE cannot assign individuals to source streams due to its difficulty in identifying discrete levels of population substructure (Waples and Gaggiotti 2006), the same six individual migrants in Beaver Creek and Lamothe Hollow had Q-values of at least 0.50 to a population different from the place where they were sampled (Table 4). Dispersal from assigned source streams to sampled streams ranged from 3.3 to 12.3 stream kilometers (Table 4).

DISCUSSION

The results of this study show that there was not a significant loss of genetic diversity in populations sampled above culverts that were classified as impassable. Our results concur with those of Torterotot et al. (2014), who also failed to detect significant losses of genetic diversity in Brook Trout populations upstream of culverts. Although individuals may not have been able to disperse between populations as a result of culverts, patch sizes may have been sufficiently large to prevent significant genetic drift (Whiteley et al. 2013). Additionally, estimates of N_e were comparable between Beaver Creek and all other sites with the exception of Lamothe Hollow. However, estimates of N_e when using low sample sizes have been shown to be biased when sample size is below the true N_e (England et al. 2006). Despite the apparent maintenance of genetic diversity in Brook Trout populations above culverts, our results indicate that Brook Trout populations located upstream of impassable culverts were more genetically differentiated relative to Brook Trout populations inhabiting streams with passable culverts or no culverts. For populations of Brook Trout inhabiting streams with a culvert classified as impassable, nearly all pairwise F_{ST} values were significant and were found to be genetically differentiated using Bayesian

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TABLE 3. Pairwise genetic differentiation index (F_{ST}) scores for Brook Trout at each sampling location as calculated by FSTAT version 2.9.3 (Goudet 2001). Pairwise F_{ST} values are above the diagonal, and P-values are below the diagonal. Bold italic values indicate significance after sequential Bonferroni correction ($P \leq 0.0008$).

Site	Buck Run Upper	Buck Run Middle	Black Run 1 Upper	Black Run 1 Middle	Upper Rocky	Lower Rocky	Oates Run	Spruce Lower	Spruce Upper	Black Run 2 Lower	Black Run 2 Upper	Lamothe Hollow	Lamothe Hollow after	Beaver Creek 1 before	Beaver Creek 1 after	Beaver Creek 2 after
Buck Run Upper		0.008	***0.038***	0.019	0.040	0.021	0.018	0.040	0.037	0.020	0.014	***0.055***	***0.071***	***0.047***	***0.051***	***0.054***
Buck Run Middle	0.0604		***0.044***	***0.013***	0.019	0.016	***0.025***	***0.022***	***0.024***	***0.017***	***0.030***	***0.038***	***0.048***	***0.065***	***0.056***	***0.078***
Black Run 1 Upper	***0.0004***	***0.0004***		0.020	***0.059***	0.015	***0.047***	0.031	0.031	***0.030***	0.028	***0.039***	***0.060***	0.023	***0.035***	***0.059***
Black Run 1 Middle	0.0042	***0.0004***	0.0038		0.029	0.000	***0.027***	0.013	0.006	0.009	0.022	***0.047***	***0.035***	0.017	0.030	***0.041***
Upper Rocky	0.0021	0.0021	***0.0004***	0.0063		0.001	***0.029***	0.010	0.028	0.015	***0.034***	***0.066***	***0.072***	***0.073***	***0.059***	***0.095***
Lower Rocky	0.0525	0.0133	0.0400	0.5371	0.0220		0.006	0.000	0.000	0.006	0.018	***0.027***	0.030	0.032	0.037	***0.059***
Oates Run	0.0483	***0.0004***	***0.0004***	***0.0004***	***0.0008***	0.0229		0.020	0.017	0.010	0.013	***0.063***	***0.073***	***0.073***	***0.074***	***0.095***
Spruce Lower	0.0022	***0.0004***	0.0013	0.0363	0.0583	0.7267	0.0033		0.000	0.000	0.020	***0.042***	***0.056***	***0.062***	***0.058***	***0.089***
Spruce Upper	0.0013	***0.0004***	0.0017	0.1513	0.0063	0.4896	***0.0004***	0.8150		0.003	0.024	***0.055***	***0.060***	***0.073***	***0.067***	***0.087***
Black Run 2 Lower	0.0317	***0.0004***	***0.0008***	0.2571	0.0067	0.0338	0.0092	0.512	0.0242		0.000	***0.053***	***0.065***	***0.054***	***0.062***	***0.090***
Black Run 2 Upper	0.0238	***0.0004***	0.0021	0.0975	***0.0008***	0.0433	0.0125	0.0704	0.0054	0.6888		***0.067***	***0.074***	***0.054***	***0.061***	***0.091***
Lamothe Hollow	***0.0004***	***0.0004***	***0.0004***	***0.0004***	***0.0004***	***0.0004***	***0.0004***	***0.0004***	***0.0004***	***0.0004***	***0.0004***		0.007	***0.064***	***0.069***	***0.095***
Lamothe Hollow after	***0.0004***	***0.0004***	***0.0004***	***0.0004***	***0.0004***	0.0038	***0.0004***	***0.0004***	***0.0004***	***0.0004***	***0.0004***	0.0125		***0.068***	***0.071***	***0.103***
Beaver Creek 1 before	***0.0004***	***0.0004***	0.0033	0.0267	***0.0004***	0.0175	***0.0004***	***0.0008***	***0.0004***	***0.0004***	***0.0004***	***0.0004***	***0.0004***		0.000	0.021
Beaver Creek 1 after	***0.0004***	***0.0004***	***0.0004***	0.1820	***0.0004***	0.0173	***0.0004***	***0.0004***	***0.0004***	***0.0004***	***0.0004***	***0.0004***	***0.0004***	0.5725		***0.036***
Beaver Creek 2 after	***0.0004***	***0.0004***	***0.0004***	***0.0004***	***0.0004***	***0.0004***	***0.0004***	***0.0004***	***0.0004***	***0.0004***	***0.0004***	***0.0004***	***0.0004***	0.0304	***0.0004***	

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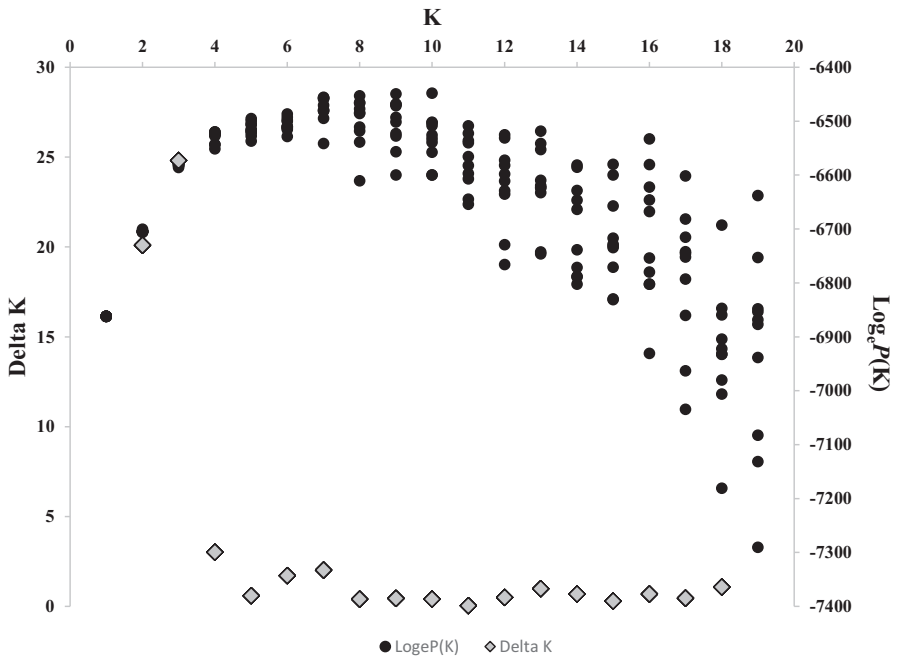

FIGURE 2. Natural log likelihood plotted against the number of possible Brook Trout populations (K) and ΔK plot. Low variability and a strong peak at a ΔK value of 3 provide support for three possible populations.

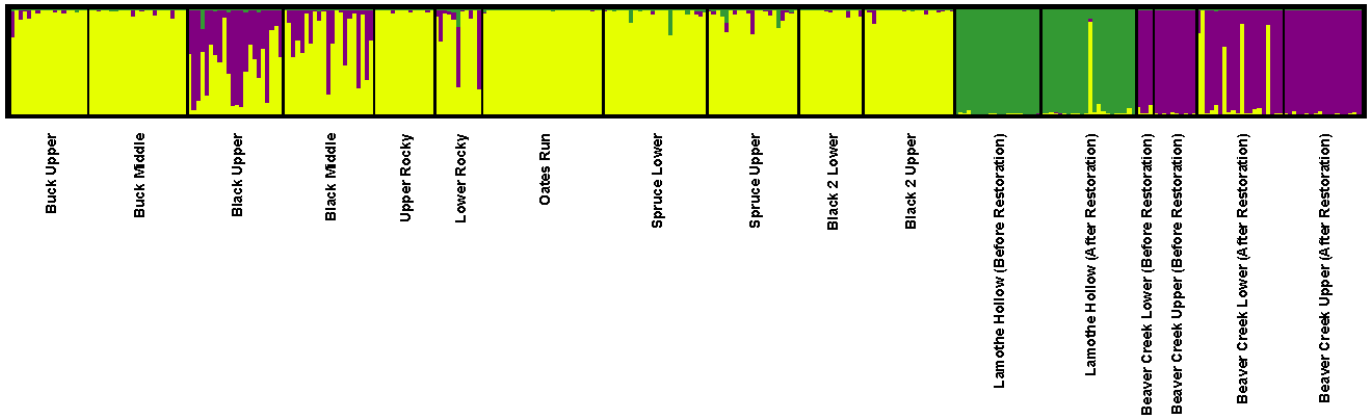

FIGURE 3. STRUCTURE plot for Brook Trout after 100,000 Markov chain–Monte Carlo repetitions using sampling locations as known priors (K = 3 populations). [Color figure can be viewed at afsjournals.org.]

modeling. However, a recently installed culvert at Oates Run, classified as impassable, did not exhibit the same differentiation as the other streams; therefore, the age of the barrier and the species' generation time should be considered when using gene flow to assess culvert passability.

Results of pairwise F_{ST} comparisons and Bayesian population structure analysis in STRUCTURE were concordant, with the exception of Buck Run. The majority of Buck Run Middle sites had significant pairwise F_{ST} comparisons with the Shavers Fork metapopulation. This result is consistent with those of Aunins et al. (2014), who found that the Buck Run population was genetically differentiated from those in other streams within Shaver's Fork and suggested that this may be due to the formation of an alluvial fan during periods of low flow or a thermal barrier in the main stem of the upper Shavers Fork during summer months. However, both analyses revealed the possibility of one-way movement between Beaver Creek (before restoration) and Black Run. Although unidirectional movements through dispersal barriers deplete upstream populations, population connectivity can be maintained through these movements (Pracheil et al.

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TABLE 4. Individual Brook Trout sampled after culvert restoration in Beaver Creek and Lamothe Hollow that were classified as first-generation migrants by using a Bayesian method (Rannala and Mountain 1997) in GENECLASS version 2.0 (Piry et al. 2004). The frequency-based method (Paetkau et al. 1995) produced the same results. The −log(L_home/L_Max) is the likelihood that an individual originated from outside its sampling location; −log(L) is the log likelihood of belonging to the assigned sourced population and significance value ($P < 0.05$). Cluster membership coefficients (Q) calculated in STRUCTURE (Pritchard et al. 2000) are listed for fish with a Q-value of at least 0.50 for a population from which the individual was not sampled. Individuals were sampled from the restored streams less than 1 year after culvert replacement.

Individual's sampling location	−log(L_home/ L_Max)	P-value	−log(L)	Q	Source stream	Distance traveled (stream km)	Length (mm)	Weight (g)
Beaver Creek	2.540	0.035	8.037	0.768	Buck Run Upper	3.3	97	13.5
	4.419	0.004	8.650	0.992	Buck Run Upper	3.3	91	12.0
	4.137	0.005	6.686	0.871	Black Run 2 Upper	12.3	116	22.7
	2.925	0.027	7.676	0.653	Black Run 1 Upper	5.5	122	23.3
	3.015	0.002	6.788	0.864	Buck Run Upper	3.3	137	40.4
Lamothe Hollow	4.163	0.000	5.817	0.889	Oates Run	4.2	87	12.2

2015). However, one-way dispersal movements were minimal enough that both Beaver Creek populations maintained differentiation. Additionally, while culverts with high outlet hang (>1 m) were likely sources of genetic differentiation, this study found that culverts with minimal (<15 cm) or no outlet hang were not barriers to gene flow within Brook Trout populations. These findings are similar to those of Torterotot et al. (2014), who reported that Brook Trout populations separated by culverts with high outlet hangs were associated with higher levels of genetic differentiation; our results also concur with Whiteley et al. (2013), who found that the genetic structure of Brook Trout was influenced by artificial barriers (i.e., dams).

After establishing impassable culverts as sources of genetic differentiation between Brook Trout populations, improvements were made to remedy high-outlet drops in two culverted stream sections that were deemed impassable. Genetic analysis indicated rapid immigration (<1 year after removal) into two previously isolated subpopulations of Brook Trout where connectivity was restored. These results concur with those of Roghair and Dolloff (2005), where migration of Brook Trout was detected within 1 year after complete removal of a Brook Trout population following a catastrophic flood. Genetic assignment of first-generation migrants revealed three populations (Black Run 1, Buck Run, and Black Run 2) as possible sources of adult migrants into Beaver Creek and indicated that Oates Run was the source of the individual migrant into Lamothe Hollow after barrier removal. Although individuals moved into both restored streams, the majority of migrants into Beaver Creek (80%) were only detected at the stream reach closest to the restored culvert (site 1). These observations are similar to those of Franklin and Bartels (2012), who noted the highest increases of fish densities in reaches closest to a restored culvert site. Black Run 1 and Buck Run are neighboring streams to Beaver Creek, and Oates Run is adjacent to Lamothe Hollow; however, Black Run 2 is over 12 stream kilometers from Beaver Creek (Table 4).

Extensive dispersal movements are not rare in the upper Shavers Fork watershed. Using radiotelemetry, Hansbarger et al. (2010) found that movements of Brook Trout in the Shavers Fork main stem were upwards of 6.5 km in a single summer. Additionally, the assignment of an individual to Oates Run indicates that downstream movement through the long (40-m), high-gradient (>3°) culvert occurs. However, the finding of a single individual migrant in Lamothe Hollow is not surprising. Lamothe Hollow (basin size = 1.3 km^2) is a much smaller stream than Beaver Creek (basin size = 4.4 km^2); however, prerestoration population surveys only observed 10 subadults or adults (>60 mm; Petty et al. 2005) in Beaver Creek 1 and 4 adults in Beaver Creek 2, whereas 25 adult individuals were sampled from Lamothe Hollow 1 year before restoration. The reduced number of individuals inhabiting Beaver Creek before restoration, accompanied by the increased number of individuals from Beaver Creek after restoration, suggests that more habitat may have become available, allowing for more colonizers or an increase in the local population. The prerestoration surveys also support the hypothesis that individuals were likely dispersing out of Beaver Creek before culvert restoration but were unable to return due to the high-outlet culvert. However, we did not detect individuals from Lamothe Hollow in other streams before culvert restoration. Surveying before culvert restoration at Lamothe Hollow detected an absence of a plunge pool and minimal stream depth below the culvert, therefore making downstream movements difficult.

Identifying the sources of first-generation migrants in our study provided an additional tool with which to characterize the dispersal ability and the dynamic behavior of

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Brook Trout in the upper Shavers Fork watershed. Although individuals were assigned to source streams through the highest likelihood identified in GeneClass2 (Piry et al. 2004), the possibility of source stream misassignment exists (type I error). However, unlike tagging studies, which may fail to detect movement when it exists (type II error), the cost of misassigning an identified migrant to an incorrect source stream is much lower (Berry et al. 2004) and provides a method to identify patterns (i.e., proximity to restored stream) of immigration in the watershed. Additionally, unlike tagging studies, which evaluate passability through re-identification of a previously marked individual, analyses including genetic assignment (Berry et al. 2004) and sibship-based methods (Whiteley et al. 2014) provide an indirect, population-based assessment tool. Although both tools can be utilized to detect movement across multiple time periods and spatial scales, the rate of admixture may reduce the effectiveness of genetic assignment as the population becomes genetically homogeneous (Berry et al. 2004), therefore necessitating methods with finer-scale resolution (e.g., sibship-based approach).

Management Implications

The results of our study suggest that genetic tools can be used in conjunction with physical assessments to evaluate and prioritize culvert restoration if such practices are available. Although the Love and Taylor (2003) phase I protocol correctly identified impassable culverts through an assessment of culvert physical characteristics, genetic analyses—including measures of population differentiation and genetic assignment—can provide a tool to both identify priority culverts for restoration and assess the success of such practices. Our study was limited to characterizing potential migrants less than 1 year after restoration, but it illustrates the utility of using genetics to detect migration after barrier removal. Through genetic assignment, we identified individual migrants from several source streams, reinforcing the importance of watershed-scale management of dispersal corridors and interconnection of habitat patches. Additionally, restoration of culverts can provide rapid success, especially within areas where dispersal is key to species' success. Although other aquatic organisms may not display the same fluvial behavior as Brook Trout in the Shavers Fork watershed, stochastic extinction events may necessitate recolonization from neighboring habitat patches for continued occupancy (Hanski 1998). Therefore, while recolonization of such species may not replicate the same temporal scale observed in our study, systems that are home to species with restricted movements should not be excluded from barrier restoration for long-term persistence of populations (Hanski 1999).

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SUPPORTING INFORMATION

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